

Relatório de Critérios de Aceite — HU #9866 · Alternador de Camadas de Infraestrutura Atual (CRAS/CREAS)

Sprint 9 · Painel de Georreferenciamento do Tenant · CA09 · Evidências end-to-end no client

Gerado em: 05/07/2026

Specs de origem: hu-geo-01-painel-georreferenciamento-evolucao.md (RN17/CA27–CA29) · HU-9863 RN09 (evidência)

Organização (tenant): E2E CadÚnico · Fortaleza · IBGE 2304400

Branch: sprint9-geo-busca-setor (client f1a3ce2 — funcionalidade entregue na HU-GEO-01 #9862; nenhum código novo nesta HU)

Ambiente: API e2e :8090 (esuas_app / esuas_e2e, PostGIS) · client build preview :4173

Ferramenta: playwright-cli (Chromium) + Pest 4

Resumo

1 critério de aceite (CA09). A funcionalidade — camada built-in "CRAS e CREAS" ligável/desligável na sidebar de camadas, plotando marcadores fixos nas coordenadas reais das unidades geocodificadas do tenant — foi entregue na HU-GEO-01 (#9862) e já revalidada na HU #9863; esta HU (#9866) é 100% evidência, sem código novo. O cenário do CA09 foi dirigido ao vivo: filtro de extrema pobreza aplicado (manchas de calor + legenda), seletor ligado (2 marcadores plotados: CRAS Fixture Centro e CREAS Fixture Leste, sobre o calor e o limite municipal) e desligado (marcadores removidos, calor preservado). Backend de suporte: endpoint tenant-scoped GET geo-panel/social-units (fetch sob demanda com cache, type_code exposto). Regressão: Pest geo-panel 51 testes / 186 asserções verdes.

CA	CRITÉRIO	STATUS
CA09	Exibição das unidades existentes sobre as manchas de calor (PO)	ATENDIDO · navegador

CA09 — Exibição das unidades existentes sobre as manchas de calor (PO)

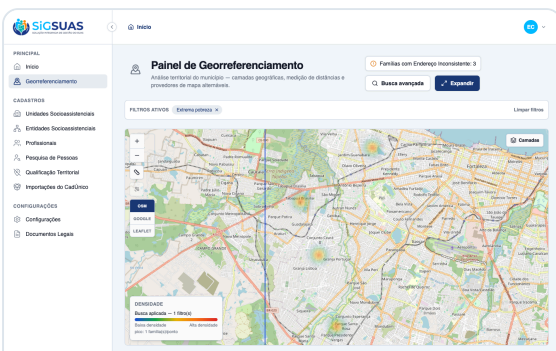
ATENDIDO · navegador

Critério (Gherkin): Dado que apliquei o filtro de extrema pobreza e visualizo as manchas de calor, Quando ativo o seletor de "Unidades de Atendimento", Então o sistema plota marcadores fixos indicando onde estão localizados os prédios físicos dos CRAS e CREAS atuais do município.

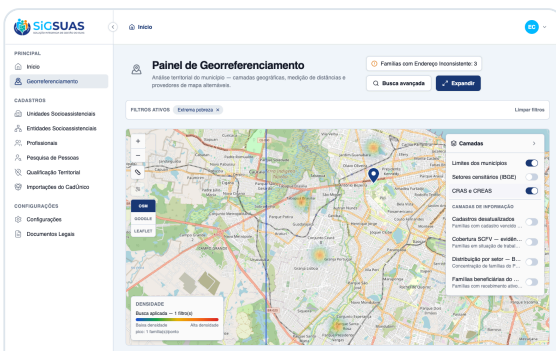
Como foi testado: Login master → /app/georreferenciamento → Busca avançada → **Extrema pobreza** → Aplicar no mapa (calor + legenda) → sidebar **Camadas** → ligar **CRAS e CREAS** → ajustar zoom para o território completo → desligar o seletor.

Resultado observado:

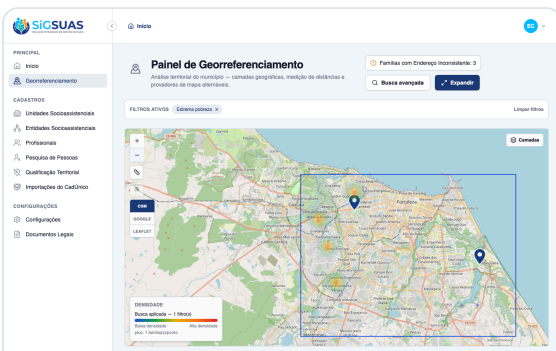
- Com o seletor DESLIGADO: apenas as manchas de calor da busca (estado de contraste).
- Com o seletor LIGADO: 2 marcadores fixos plotados nas coordenadas reais das unidades do município — CRAS Fixture Centro (noroeste) e CREAS Fixture Leste (sudeste) — sobrepostos às manchas de adensamento e ao limite municipal, permitindo confrontar visualmente serviço instalado × risco.
- Com o seletor DESLIGADO novamente: marcadores removidos, calor e legenda preservados (alternância sem recarregar o painel).
- Isolamento de tenant: unidades servidas por GET geo-panel/social-units sob EnsureTenantScope (cobertura na suíte Pest do apêndice).



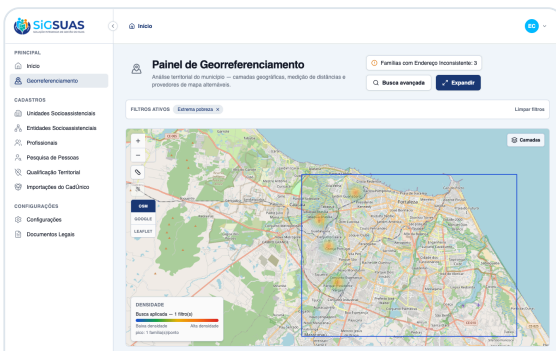
ca09-calor-sem-marcadores.png



ca09-toggle-camada-ligado.png



ca09-marcadores-sobre-calor.png



ca09-toggle-desligado.png

Anexo — Cobertura automatizada (Pest, --filter=GeoPanel — 51 testes / 186 asserções, inclui endpoint social-units e isolamento tenant)

```
PASS Tests\Unit\Services\Geo\GeoPanelQueryServiceTest
✓ it filters extreme_poverty by per_capita_income against the configu... 0.32s
```

- ✓ it filters single_person as exactly one ACTIVE member 0.03s
- ✓ it filters receives_pbf on the family flag 0.01s
- ✓ it filters member flags (child_labor, homeless_situation, has_disab... 0.05s
- ✓ it filters indigenous_quilombola as an OR between the two family fl... 0.02s
- ✓ it filters gpte by any traditional group, refinable by traditional_... 0.02s
- ✓ it intersects multiple criteria with AND (RN06) 0.03s
- ✓ it classifies expired as more than 24 months since source_updated_a... 0.02s
- ✓ it classifies expiring as between 19 and 24 months since source_upd... 0.02s
- ✓ it restricts by sector_uuid (SQLite fallback: sector city = family... 0.02s
- ✓ it aggregates by sector, groups anonymous points and computes the b... 0.02s
- ✓ it ignores families without an active locatable address point 0.02s
- ✓ it scopes every aggregate to the given tenant (RN01) 0.02s
- ✓ it validates that a sector belongs to the tenant municipality (RN08... 0.01s

PASS Tests\Feature\Api\Client\GeoPanelExportTest

- ✓ it creates a pending export with the immutable filters snapshot and... 0.06s
- ✓ it blocks the export with 422 and the guidance message when no effe... 0.02s
- ✓ it requires the map image for pdf exports and stores it under the t... 0.03s
- ✓ it rejects a map image that is not a base64 png 0.02s
- ✓ it lists and shows only the exports of the active tenant, 404 on cr... 0.03s
- ✓ it downloads a completed export and refuses one still pending 0.03s
- ✓ it generates the xlsx from the filters snapshot with the tenant pre... 0.04s
- ✓ it generates the pdf composing the map image sent by the client (CA... 0.13s
- ✓ it marks the export as failed with the error message when the job f... 0.02s
- ✓ it neutralizes formula injection on every text cell of the spreadsh... 0.01s

PASS Tests\Feature\Api\Client\GeoPanelLayerDataTest

- ✓ it merges a globally published layer after the builtins, keeping th... 0.02s
- ✓ it shows a selected-audience layer only to the covered tenant (CA43... 0.02s
- ✓ it never lists draft or inactive layers on the tenant panel (CA48/R... 0.02s
- ✓ it reflects audience/status changes on the very next fetch – no cac... 0.02s
- ✓ it executes the layer criteria strictly under each consumer tenant... 0.03s
- ✓ it answers 404 (never 403) for a layer that is not visible to the t... 0.03s
- ✓ it serves the CURRENT definition of the layer – an admin edit refle... 0.03s
- ✓ it shapes the payload by representation: markers, boundaries and he... 0.03s
- ✓ it denies the layer data endpoint to an operator without the geo_pa... 0.02s

PASS Tests\Feature\Api\Client\GeoPanelTest

- ✓ it returns the tenant municipality with geometry, centroid and the... 0.04s
- ✓ it responds 409 geo_panel.city_unavailable when the tenant has no r... 0.02s
- ✓ it responds 409 when the municipality geometry exists but is inacti... 0.02s
- ✓ it lists exactly the 3 builtin layers with the stable contract (CA2... 0.02s
- ✓ it serves the census sectors of the tenant municipality as a GeoJSO... 0.02s
- ✓ it serves the tenant social units as anonymous points with type_cod... 0.02s
- ✓ it applies the AND intersection of combined criteria scoped to the... 0.03s
- ✓ it never leaks families from another tenant in the search (RN01/CA3... 0.02s
- ✓ it returns an empty payload when no record matches, so the map stay... 0.02s
- ✓ it rejects a search without any effective criterion (Seção 10 – reg... 0.02s
- ✓ it rejects a sector_uuid that belongs to another municipality (RN08... 0.02s
- ✓ it always includes the fixed RN07 indicator in the summary, even wi... 0.03s
- ✓ it aggregates the summary by census sector under the active filters... 0.02s
- ✓ it denies every geo-panel endpoint to the base operator without the... 0.02s
- ✓ it grants the full panel to the operator once the geo_panel_viewer... 0.03s
- ✓ it grants the full panel to the Master without any add-on role (RN2... 0.02s
- ✓ it does not let the geo_panel_viewer role in Tenant A open the pane... 0.02s
- ✓ it exposes geo_panel_viewer as an add-on role, never a hierarchy po... 0.02s

Tests: 51 passed (186 assertions)

Duration: 1.87s

